

HAT-P-12b

R-0720

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_250327 · created 2026-07-27T23:52:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460761.7801 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.039 +/- 0.05
Transit depth (Rp/Rs)^2	0.16 +/- 0.39 [%]
Orbital Inclination (inc)	85.32 +/- 1.77
Airmass coefficient 1 (a1)	1.213 +/- 0.071
Airmass coefficient 2 (a2)	-0.18 +/- 0.1
Scatter in the residuals of the lightcurve fit is	5.83 %
Best Comparison Star	#2 - [295, 274]
Optimal Aperture	1.71
Optimal Annulus	6.93
Transit Duration (day)	0.035 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.039 ± 0.05 vs 0.1406 (deviation 2.03σ)
Duration fit vs published	0.035 d vs 0.0975 d (deviation 1.89σ)
Mid-time O–C	11.1 min
Depth SNR	0.8
Residual scatter	5.83 %

Light curve

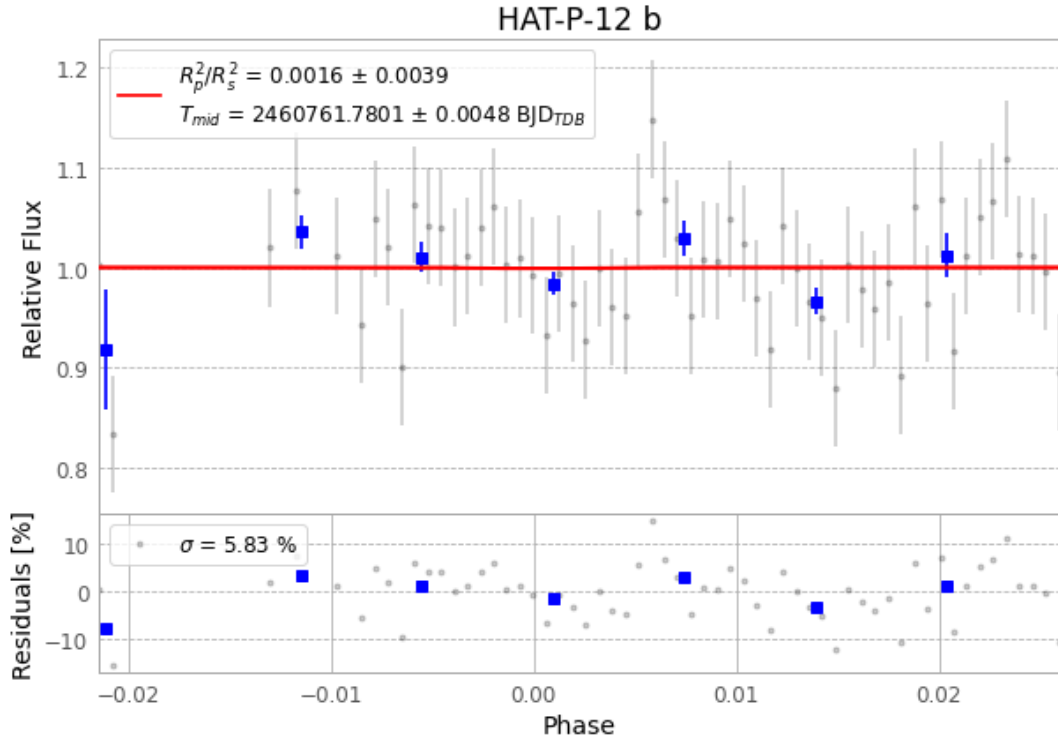


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-03-26.png (SHA-256 9aea4354ea21668d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ec6971fb41154a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), HATP-12250327045715.FITS → HATP-12250327083616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250327.log (a1a56de2754c...), FinalLightCurve_HAT-P-12 b_2025-03-26.png (9aea4354ea21...), FinalLightCurve_HAT-P-12 b_2025-03-26.csv (bc97c5989591...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:52Z.